

Name_____

Statistics 1040: Introduction to Statistics

Section 12

Test 2 – March 29, 2000

Version 1

75 minutes; closed book, closed notes, calculators and provided formula sheet allowed. 15 questions, 7 points each (max score is 105). For numeric questions, show your work and circle your answers. For essay questions, answer the questions in complete sentences. Good luck!

1. California has a population a little over 15 times as large as Utah's (33.9 million to 2.2 million). Suppose we are happy with the accuracy of a political survey of 1,500 Californians. If we wish to get similar accuracy in Utah, about how many Utahns should we survey? Briefly explain.
2. I draw 2 cards without replacement from a well-shuffled standard, 52-card deck, which contains 26 black cards (spades and clubs) and 26 red ones (hearts and diamonds). What is the chance that I will draw at least one black card?
3. To gauge USU student support for President Hall's proposed 9% tuition increase, the ASUSU hire a student to conduct a survey. The pollster goes door-to-door in the family housing sections of Aggie Village, asking the students who come to the door, "Do you believe that USU should increase tuition for working students by at least 9% for the coming year?"

Name at least two potential problems with this survey.

4. I shuffle a standard, 52-card deck. What is the chance that at least one of “the top card is the queen of hearts” or “the bottom card is the queen of hearts” is true?

5. A university’s college of science has 436 graduating seniors. A simple random sample of 225 of these students is polled, and 83 (37%) report plans to attend graduate school in the next year. By about how much is our estimate that 37% of the graduates of this college plan to attend graduate school next year likely to be off?

6. Using the survey results of problem 5, construct a 95% confidence interval for the population percentage who plan to attend grad school.

7. True or false and explain: There is a 95% chance that the true population percentage will fall into the interval calculated in problem 6.

8. Recall, a roulette wheel has 38 pockets, which should all be equally likely to come up. A “four numbers” bet has 4 chances to win, and pays 8 to 1. That is, if you make a \$1 “four numbers” bet, and one of your 4 numbers comes up, you keep your dollar and get an extra \$8. If any other pocket comes up, you lose your dollar.

Suppose you make 25 \$1 “four numbers” bets. Construct a box model for your winnings. Be complete!

9. What are the expected value and standard error for your winnings from the 25 \$1 “four numbers” bets in problem 8?

10. What is the chance you will be ahead (have winnings of more than \$0), after your 25 \$1 “four numbers” bets in problem 8?

11. What do we mean by “nonresponse bias” in surveys and polls? Why is it important, and what should we do to minimize its impact?

12. I roll a pair of fair (six-sided) dice, one white and one black. Let event $A =$ “The white die shows a 1.”
- (a) Give an example of an event B which is mutually exclusive of event A .
- (b) Give an example of an event C which is independent of event A .
13. A fair coin is tossed many times. Which is more likely, 20 to 30 heads in 50 tosses, or 245 to 255 heads in 500 tosses?
14. A fair six-sided die is rolled 30 times, and we record the sum of the 30 rolls. We then start over and repeat the process, getting the sum of another 30 rolls. We do this 1,000 times, and construct the histogram of the 1,000 sums. Will the histogram be “basically flat,” “a lot like the normal curve,” or some other shape entirely? Briefly explain.
15. I flip a fair coin 4 times. What is the chance that all 4 flips will come up heads?